

Bounds on Codes

We now look at the problem of determining how the parameters n, k, δ of a linear code are constrained. Given values for two of these three parameters, what range of values is possible for the third? This problem has not been solved in general, although it has been settled for many small values of n, k and δ .

5.1.1 Lemma [Spoiling Lemma] If there exists a linear (n, k, δ) code, then there exists a linear (n, k, d) code for $d = 1, 2, \dots, \delta$.

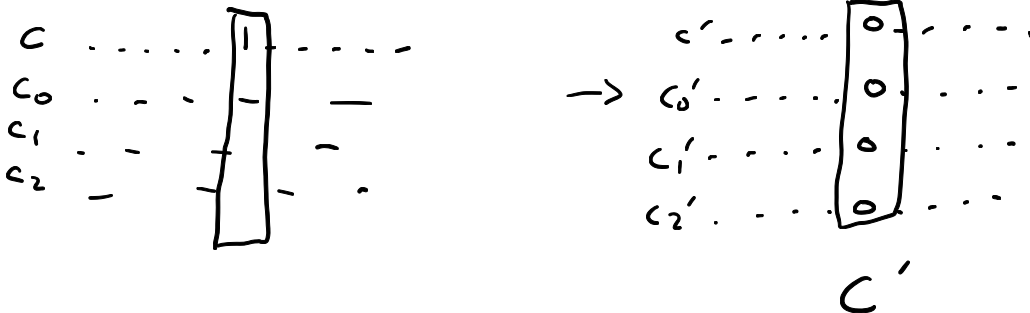
Proof sketch

The case $\delta = 1$ is trivial.

Assume $\delta > 1$, and suppose $\exists$ a linear (n, k, δ) code C .

$\exists c \in C$ s.t. $\|c\| = \delta$

Pick a position in which c has a 1 and make this position 0 in all the codewords of C , to create a new code C' .



Prove that C' is a linear $(n, k, \delta-1)$ code.

Repeat the process to prove the lemma.

- C' has length n
- C' is linear
- C' has dimension k
- $\delta_{C'} = \delta_C - 1$

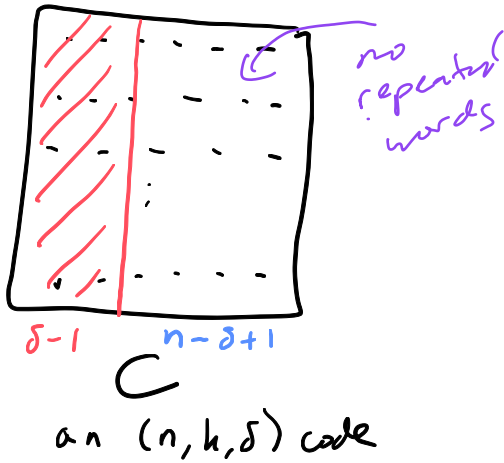
5.1.2 Theorem [Singleton Bound] In any code (linear or not), $|C| \leq 2^{n-\delta+1}$. In particular, if $|C| = 2^k$, then

(5.1.1)

$$n \geq k + \delta - 1.$$

Proof sketch

Delete the first $\delta-1$ bits of each codeword.



Since C has distance δ the resulting words of length $n-\delta+1$ will be distinct. There are only $2^{n-\delta+1}$ words of length $n-\delta+1$ so

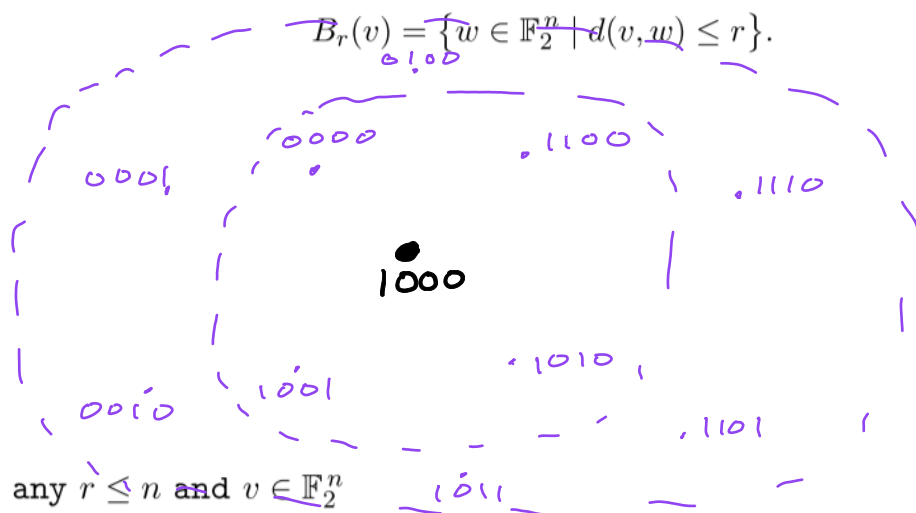
$$|C| \leq 2^{n-\delta+1}.$$

5.2. The Hamming Bound

5.2.1 Definition Let $v \in \mathbb{F}_2^n$ and let $r \geq 0$ be an integer. The (closed) ball of radius r centred at v is the set

Example

$B_2(1000)$



5.2.2 Lemma For any $r \leq n$ and $v \in \mathbb{F}_2^n$

$$|B_r(v)| = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{r}.$$

words of length n that differ from v in i places is $\binom{n}{i}$.

5.2.3 Theorem [Hamming Bound] If C is a code of length n (linear or not) and distance δ then

$$|C| \leq \frac{2^n}{\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{d}} \quad \text{where } d = \left\lfloor \frac{\delta - 1}{2} \right\rfloor.$$

Every two codewords are at distance $\geq \delta$ and $d = \left\lfloor \frac{\delta - 1}{2} \right\rfloor$ so the balls $B_d(c)$ do not overlap for any codewords.

So $|C| \left(\underbrace{\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{d}}_{\substack{\text{\# words of distance} \\ \text{at most } d \text{ from a} \\ \text{codeword}}} \right) \leq \text{total \# of words of length } n = 2^n$

If C is linear so $|C| = 2^k$

$$2^k \left(\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{d} \right) \leq 2^n$$

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{d} \leq 2^{n-k}$$

$$\log_2 \left(\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{d} \right) \leq n - k$$

Exercise Determine the lower bounds on the length of a 2-error correcting code with 64 codewords, given by the Singleton and Hamming bounds.

$$|C| \leq 2^{n-d+1} \quad |C| \leq \frac{2^n}{\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{d}} \quad \text{where } d = \left\lfloor \frac{d-1}{2} \right\rfloor$$

Singleton

$$d \geq 5 \quad 64 \leq 2^{n-4} \quad n \geq 10$$

Hamming

$$d=5 \quad d=2 \quad |C| \leq \frac{2^n}{\binom{n}{0} + \binom{n}{1} + \binom{n}{2}} \quad 2^6 \leq \frac{2^n}{1+n+\frac{n(n-1)}{2}}$$

$$n^2 + n + 2 \leq 2^{n-5} \quad n \geq 13$$

In fact, we need $n \geq 14$.

5.2.6 Example Consider the Hamming bound for the $(7, 4, 3)$ code C with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

which was discussed at the end of the Section 4.11. We have $n = 7$, $\delta = 3$. Thus

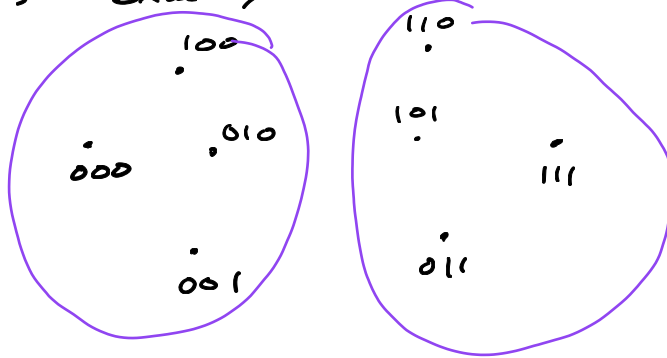
$$|C| \leq \frac{2^7}{\binom{7}{0} + \binom{7}{1}} = \frac{128}{8} = 16.$$

And in fact $|C| = 16$, so the bound is sharp for this code. This means $\mathbb{F}_2^7$ is exactly covered by the balls of radius 1 centred on the 16 words of C .

Similarly for $C = \{000, 111\}$ $\delta = 3$ so $d = 1$

$$|C| = 2 = \frac{2^3}{\binom{3}{0} + \binom{3}{1}} = \frac{2^n}{\binom{n}{0} + \binom{n}{1}}$$

$\mathbb{F}_2^3$ is exactly covered by balls of radius 1.



5.3. The Griesmer Bound

5.3.1 Theorem [Griesmer Bound] Let C be a linear (n, k, δ) code. Then

$$(5.3.1) \quad n \geq \sum_{j=0}^{k-1} \left\lceil \frac{\delta}{2^j} \right\rceil.$$

Exercise
What is the lower bound on the length of a 2-error correcting code with 64 codewords given by the Griesmer bound? $k=6$ $\delta \geq 5$ so use $\delta=5$

$$\begin{aligned} n &\geq \sum_{j=0}^5 \left\lceil \frac{5}{2^j} \right\rceil = \left\lceil \frac{5}{1} \right\rceil + \left\lceil \frac{5}{2} \right\rceil + \left\lceil \frac{5}{4} \right\rceil + \left\lceil \frac{5}{8} \right\rceil + \left\lceil \frac{5}{16} \right\rceil + \left\lceil \frac{5}{32} \right\rceil \\ &= 5 + 3 + 2 + 1 + 1 + 1 \\ &= 13 \end{aligned}$$

5.3.1 Theorem [Griesmer Bound] Let C be a linear (n, k, δ) code. Then

$$(5.3.1) \quad n \geq \sum_{j=0}^{k-1} \left\lceil \frac{\delta}{2^j} \right\rceil.$$

Proof Let $L(k, \delta)$ denote the minimum length of a linear code of dimension k and distance δ .

Step ①: Prove that $L(k, \delta) \geq L(k-1, \lceil \frac{\delta}{2} \rceil) + \delta$.

Step ②: Use induction on k to prove that

$$n \geq \sum_{j=0}^{k-1} \left\lceil \frac{\delta}{2^j} \right\rceil \quad (\text{using the equation from Step ①})$$

Base case $k=1$
$$\sum_{j=0}^0 \left\lceil \frac{\delta}{2^j} \right\rceil = \left\lceil \frac{\delta}{2^0} \right\rceil = \delta$$

Clearly $n \geq \delta$ so base case is true.

Inductive hypothesis: Eq 5.3.1 holds for $k-1$, that is,

$$n \geq \sum_{j=0}^{k-2} \left\lceil \frac{\delta}{2^j} \right\rceil$$

Now prove Eq 5.3.1 for k .

$$n \geq L(k, \delta)$$

Definition of $L(k, \delta)$

$$\geq L(k-1, \lceil \delta/2 \rceil) + \delta$$

Equation 5.3.5

$$\geq \sum_{j=0}^{k-2} \left\lceil \frac{\lceil \delta/2 \rceil}{2^j} \right\rceil + \delta$$

Inductive hypothesis $k-1$

$$= \sum_{j=0}^{k-2} \left\lceil \frac{\delta}{2^{j+1}} \right\rceil + \delta$$

Lemma 2.3.7

$$\left\lceil \frac{\lceil \frac{x}{m} \rceil}{n} \right\rceil = \left\lceil \frac{x}{mn} \right\rceil$$

$$= \sum_{i=0}^{k-1} \left\lceil \frac{\delta}{2^i} \right\rceil$$

Put $i = j+1$.

$$\left\lceil \frac{\delta}{2^1} \right\rceil + \left\lceil \frac{\delta}{2^2} \right\rceil + \dots + \left\lceil \frac{\delta}{2^{k-1}} \right\rceil$$

Prove that $L(k, \delta) \geq L(k-1, \lceil \frac{\delta}{2} \rceil) + \delta$.

Fix k and δ . Let C be a linear code of dimension k and distance δ of minimal length $n = L(k, \delta)$. By Theorem 4.2.1 there is a word $c \in C$ of exact weight δ , and no smaller weight non-zero word. Since equivalent codes have the same parameters, wlog let $c = (\underbrace{11 \dots 1}_{\delta} \mid \underbrace{00 \dots 0}_{n-\delta}) = (\mathbf{1}_{\delta} \mid \mathbf{0}_{n-\delta})$.

Extend $\{c\}$ to a basis for C . Let G be a corresponding generator matrix for this basis, with first row c . Thus C has a generating matrix

$$G = \left(\begin{array}{ccc|ccc} 1 & \dots & 1 & 0 & \dots & 0 \\ * & * & * & & & \\ \vdots & \vdots & \vdots & & & \end{array} \right) \quad \begin{array}{l} \delta \quad n-\delta \\ \boxed{G'} \text{ is } (k-1) \times (n-\delta) \end{array}$$

First prove that the rows of G' are linearly independent.

$$\left(\begin{array}{ccc|ccc} 1 & \dots & 1 & 0 & \dots & 0 \\ g_1 & & & g'_1 & & \\ g_2 & & & g'_2 & & \\ \vdots & & & \vdots & & \\ g_{k-1} & & & g'_{k-1} & & \end{array} \right)$$

Suppose the rows of G' are not linearly independent, so $\exists$ a subset of $\{g'_1, g'_2, \dots, g'_{k-1}\}$ that sum to $\mathbf{0}$.

For example, suppose $g'_1 + g'_2 + g'_3 = \mathbf{0}$.

Then $(g_1 \mid g'_1) + (g_2 \mid g'_2) + (g_3 \mid g'_3) = (g_1 + g_2 + g_3 \mid \mathbf{0} \dots \mathbf{0})$.

Now $(g_1 + g_2 + g_3 \mid \mathbf{0} \dots \mathbf{0}) \in C$ so $\|g_1 + g_2 + g_3\| \leq \delta$.

If $\|g_1 + g_2 + g_3\| < \delta$ then contradiction because this gives a codeword of non-zero weight $< \delta$.

So $\|g_1 + g_2 + g_3\| = \delta$ then $g_1 + g_2 + g_3 = \mathbf{0}$ so G contains a linearly dependent set $\Rightarrow \Leftarrow$.

Now consider the linear code $C' = f(C)$ generated by G' . $n' = n - \delta$ $k' = k - 1$

Let $w \in f(C)$ be a non-zero word of minimal weight. So there exists $u \in C$ with $u = (\underbrace{v}_{\delta} \mid \underbrace{w}_{n-\delta})$ and $\|w\| = \delta'$. Then

$$\delta \leq \|u + c\| \quad \text{[11...1 00...0]}$$

$$u, c \in C, u \neq c \Rightarrow 0 \neq u + c \in C$$

$$= \|(v \mid w) + (\mathbf{1}_{\delta} \mid \mathbf{0}_{n-\delta})\|$$

$$= \|(v + \mathbf{1}_{\delta}) \mid (w + \mathbf{0}_{n-\delta})\|$$

$$= \|v + \mathbf{1}_{\delta}\| + \|w + \mathbf{0}_{n-\delta}\|$$

$$= \delta - \|v\| + \|w\|$$

Equation 2.2.2

$$\leq \|u\| - \|v\| + \|w\|$$

$$0 \neq u \in C \Rightarrow \|u\| \geq \delta$$

$$= \|v\| + \|w\| - \|v\| + \|w\|$$

$$u = (v \mid w)$$

$$= 2\|w\| = 2\delta'.$$

$$L(k, \delta) - \delta = n - \delta \geq L(k-1, \delta') \geq L(k-1, \lceil \frac{\delta}{2} \rceil) \quad \delta \leq 2\delta' \quad \delta' \geq \lceil \frac{\delta}{2} \rceil$$

5.1.2 Theorem [Singleton Bound] In any code (linear or not), $|C| \leq 2^{n-\delta+1}$. In particular, if $|C| = 2^k$, then

$$(5.1.1) \quad n \geq k + \delta - 1.$$

5.2.3 Theorem [Hamming Bound] If C is a code of length n (linear or not) and distance δ then

$$|C| \leq \frac{2^n}{\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{d}} \quad \text{where } d = \left\lfloor \frac{\delta-1}{2} \right\rfloor.$$

5.3.1 Theorem [Griesmer Bound] Let C be a linear (n, k, δ) code. Then

$$(5.3.1) \quad n \geq \sum_{j=0}^{k-1} \left\lceil \frac{\delta}{2^j} \right\rceil.$$

Exercise: What do each of the Singleton, Hamming and Griesmer bounds tell us about the maximum number of codewords in a linear code with length $n=20$ and distance $\delta=5$? $|C| = 2^k$

Singleton $k \leq n - \delta + 1 \quad k \leq 16$

Hamming $|C| \leq \frac{2^{20}}{\binom{20}{0} + \binom{20}{1} + \binom{20}{2}} = 4969$ C is linear so $|C|$ must be a power of 2

$$|C| \leq 4096 = 2^{12}$$

so $k \leq 12$

Griesmer $20 \geq \left\lceil \frac{5}{2^0} \right\rceil + \left\lceil \frac{5}{2^1} \right\rceil + \left\lceil \frac{5}{2^2} \right\rceil + \left\lceil \frac{5}{2^3} \right\rceil + \dots + \left\lceil \frac{5}{2^{k-1}} \right\rceil$

$$20 \geq 5 + 3 + 2 + 1 + \dots + 1$$

$\underbrace{\hspace{10em}}_{k-3}$

$$20 \geq 7 + k$$

$$k \leq 13$$



5.4. The Gilbert-Varshamov Bound

5.4.1 Theorem [Gilbert-Varshamov (G-V)] Let $n, k, \delta \in \mathbb{N}$, with $n > \delta \geq 2$. If

$$(5.4.1) \quad \binom{n-1}{0} + \binom{n-1}{1} + \dots + \binom{n-1}{\delta-2} < 2^{n-k}$$

then there exists a linear (n, k, δ) -code.

p.c. matrix $\left(\begin{array}{c} \\ \\ \end{array} \right)_{n \times (n-k)}$ l.i. cols.

Proof Suppose that the integers n, k and δ satisfy 5.4.1. We will construct an $n \times (n-k)$ matrix H with linearly independent columns and in which no set of $\delta-1$ or fewer rows is linearly dependent. This matrix is a parity check matrix for a linear (n, k, d) -code for some $d \geq \delta$, by Theorem 4.9.1. Thus, by the Spoiling Lemma 5.1.1, there exists a linear (n, k, δ) -code.

Start building the matrix H by choosing the first $n-k$ rows of H to be the rows of the matrix I_{n-k} . This guarantees that the columns of H are linearly independent. Furthermore, we are guaranteed that so far, we have $n-k$ rows, no set of $\delta-1$ of which are linearly dependent.

$$H = \left(\begin{array}{c} I_{n-k} \\ \equiv \\ \equiv \\ \equiv \end{array} \right)_{n \times (n-k)}$$

Need

- linearly independent columns
- no set of $\delta-1$ or fewer rows is linearly dependent.

Build H row by row, checking at each stage that no set of $\delta-1$ or fewer rows is lin. dep.

Given i rows, we can find a vector of length $n-k$ to put in row $i+1$ provided:

forbidden $\left(\binom{i}{0} + \binom{i}{1} + \binom{i}{2} + \dots + \binom{i}{\delta-2} \right) < 2^{n-k}$

row of all zeros

any row that is already there

the sum of 2 rows that are there

the sum of $\delta-2$ rows that are already there

To add the final row, we need

$$\binom{n-1}{0} + \binom{n-1}{1} + \dots + \binom{n-1}{\delta-2} < 2^{n-k}$$

Exercise: $\binom{n-1}{0} + \binom{n-1}{1} + \dots + \binom{n-1}{s-2} < 2^{n-k}$

(a) Show that the G-V bound guarantees the existence of a linear $(8, 3, 4)$ -code.

(b) Use the construction technique of the proof of the G-V bound to find a parity check matrix for a linear $(8, 3, 4)$ -code.

(c) From the p.c. matrix, find a corresponding generating matrix for your linear $(8, 3, 4)$ -code.

(a) $\binom{7}{0} + \binom{7}{1} + \binom{7}{2} = 29$ $2^5 = 32$ $29 < 32$ so G.V. says the code exists.

(b) $H = \begin{pmatrix} 10000 \\ 01000 \\ 00100 \\ 00010 \\ 00001 \\ 11111 \\ 11100 \\ 10011 \end{pmatrix}$ is a p.c. for a linear $(8, 3, 4)$ -code.

(c) $G = (I \ X)$ $H = \begin{pmatrix} X \\ I \end{pmatrix}$

$H' = \begin{pmatrix} 11111 \\ 11100 \\ 10011 \\ I_5 \end{pmatrix}$ $G' = \begin{pmatrix} I_3 & 11111 \\ & 11100 \\ & 10011 \end{pmatrix}$ $G = \begin{pmatrix} 11111 & 100 \\ 11100 & 010 \\ 10011 & 001 \end{pmatrix}$

5.4.2 Corollary If $n > \delta \geq 2$ then there exists a linear code C with length n and distance δ with

$$(5.4.3) \quad |C| \geq \frac{2^{n-1}}{\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2}}.$$

Proof The G-V bound guarantees the existence of a linear (n, k, δ) -code provided

$$\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2} < 2^{n-k}.$$

That is, if

$$(5.4.4) \quad 2^k < \frac{2^n}{\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2}}.$$

Let k' be the largest value of k for which inequality 5.4.4 is satisfied, so (by G-V) there does exist a linear (n, k', δ) code, C . We show that C satisfies inequality 5.4.3.

Inequality 5.4.4 is not satisfied for $k' + 1$, so

$$2^{k'+1} \geq \frac{2^n}{\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2}}$$

$$\therefore |C| = 2^{k'} \geq \frac{2^{n-1}}{\binom{n-1}{0} + \binom{n-1}{1} + \cdots + \binom{n-1}{\delta-2}}. \quad \square$$

Exercise: What does the G-V. bound tell us about the number of codewords in a linear code of length $n=20$ and distance $\delta=5$?

$\exists$ a linear $(20, k, 5)$ code with

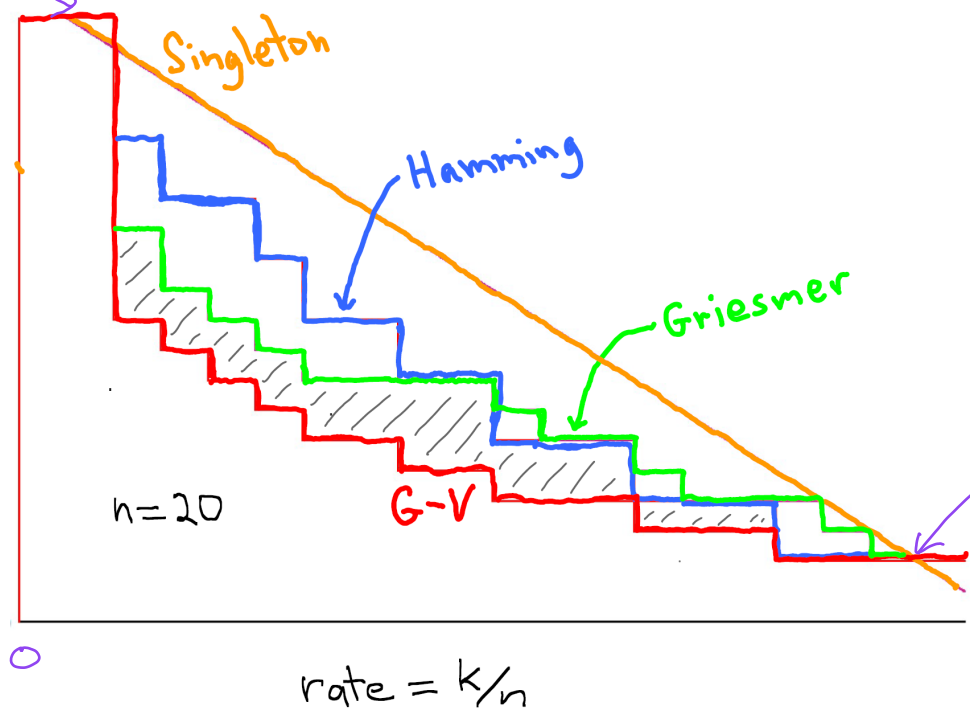
$$|C| \geq \frac{2^{19}}{\binom{19}{0} + \binom{19}{1} + \binom{19}{2} + \binom{19}{3}} = 451$$

Since linear $|C| \geq 512 = 2^9$

$(20, 9, 5) \checkmark$
 $(20, 10, 5) ?$
 $(20, 11, 5) ?$
 $(20, 12, 5) ?$
 $(20, 13, 5) \times$

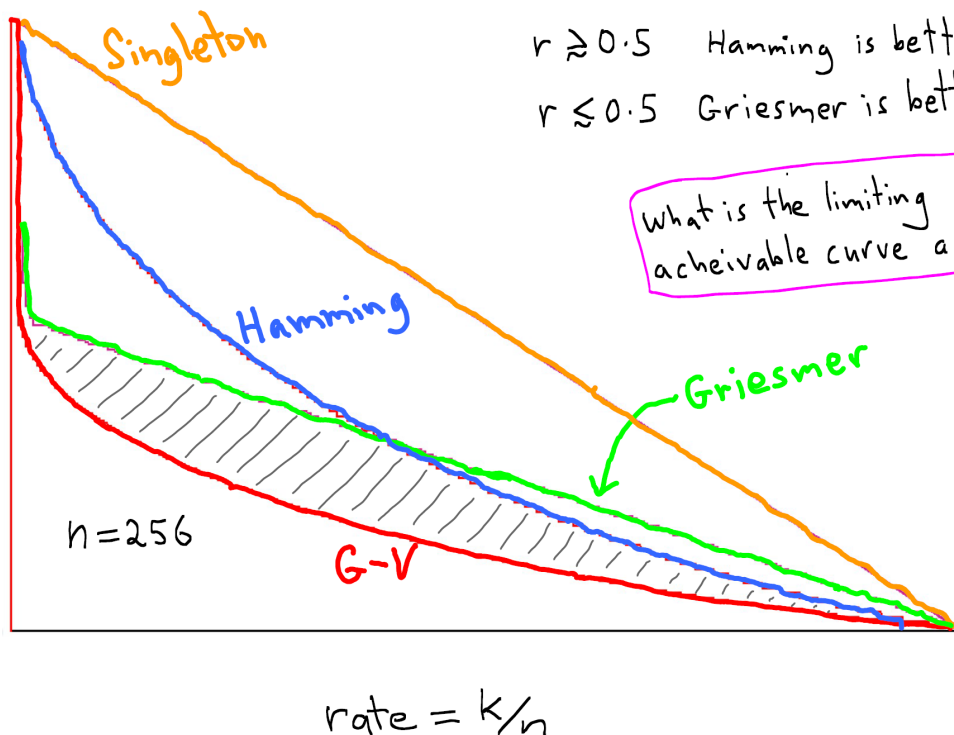
$\{0,1\}^n$ $n=20$
 $\delta=20$
 $k=1$

relative distance
 $\delta = \frac{\delta}{n}$



$C = \mathbb{F}_2^{20}$
 $n=20$
 $\delta=1$
 $k=20$

relative distance
 $\delta = \frac{\delta}{n}$



$r \geq 0.5$ Hamming is better bound.
 $r \leq 0.5$ Griesmer is better bound.

What is the limiting achievable curve as $n \rightarrow \infty$?